

PROJECTS

Verilog Projects (tools: EDA playground, Vivado)

Beginner:

1. Implement Calculator – add, subtract, multiply, divide - Gate level modelling

Medium:

1. Code converter – <http://Inkiy.in/codeconverterpdf>
2. Design a Vending machine

Advance:

1. RISC V – implementation of 5 staged pipelined RISC-V processor. Also implement hardware detection if possible.
2. Synchronous FIFO (Read about asynchronous FIFO as well) – design a synchronous FIFO.

Implement Test bench for the same and try to cover all the scenarios.

Embedded projects (tools: arduino, raspberry pi, etc)

Beginner:

1. Smart street lights using IoT – motion sensor street lights
2. Traffic light control system – railway crossing traffic light control
3. Smoke detector – sensor beeps when smoke is detected
4. Vehicle speed check and accident-avoidance system – check vehicle distance from the front vehicle continuously and buzz the buzzer if the distance is less than the set values. Mainly on highways to avoid crash.

Medium:

1. Robotic Arm - allows controlling a robotic arm by hand movements

Advance:

1. Balancing bot – this will require Control system, signal processing and embedded programming.

CMOS projects (tool: LT Spice)

Beginner:

1. Implement nmos and pmos and plot the below characteristics:
 - a. I_d vs V_{gs}
 - b. I_d vs V_{ds}
 - c. Voltage transfer characteristics.

Choose right resistance and analyse based on changing the resistance of the circuit.

Medium:

1. Implement same for CMOS as above

Advance

1. Full adder in LT-spice – implement a full adder using CMOS in LT-spice. Make your own CMOS and convert it into a lib and use it to make a full adder. That full adder can also be made into a lib and used in some other circuit.

Physical Design Project (tool: open source tool - open road)

Beginner:

1. Take simple nand gate and implement its layout.

Advance:

1. RiscV_processor.v to gds

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